Site the 2027 solar pilot at Corvane Gorsefield: its array would have avoided 0.89 t of grid carbon dioxide for every MWh it generated in 2025

To: Capital Committee
From: Barbara Rangel, Director, Energy and Sustainability
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Re: Pilot site under the 2027 On-Site Solar Pilot siting charter (minute CC-26-41)

I recommend that the Committee name Corvane Gorsefield (store 2291) as the 2027 pilot store in the capital plan. On the charter's measure, the carbon dioxide the generators of the store's balancing area would not have emitted in 2025 had the array been supplying the store, its array avoids 0.89 t per MWh it generates. Corvane Halvane is the first store left out at 0.58 t per MWh, 0.31 below. The other six arrays would each have avoided between 0.39 and 0.41 t per MWh.

An array's output is taken off the generators at the margin in the hours it generates, so what it avoids is their emission rate in its own hours. In Gorsefield's area gas carries the nights and the coal fleet follows the load from 5 am to 6 pm, the hours the array generates. Brackwell has the highest average output emission rate, 0.85 t per MWh, but by day its coal runs at full output and gas follows the load, so its array avoids 0.40. Quillbury's high non-baseload rate (1.11) comes from old coal units run on winter evenings, after sunset (0.39 in the array's hours). Tennaford's coal follows the load through the night, which puts a marginal rate fitted to every hour of 2025 above every other area's, but gas follows its business day from 7 am to 6 pm (0.41 in the array's hours).

Store	Balancing area	Array output 2025 (MWh)	Grid CO2 avoided (t/MWh)	Metered use 2025 (MWh)	Share of use supplied	Paid per MWh ($)
Corvane Gorsefield (2291)	**Gorsefield Balancing Authority**	**733**	**0.89**	**2,570**	**28.5%**	**146**
Corvane Halvane (2449)	Halvane Electric Balancing Authority	679	0.58	2,507	27.1%	142
Corvane Tennaford (2238)	Tennaford Power Pool	664	0.41	2,458	27.0%	137
Corvane Brackwell (2104)	Brackwell Balancing Authority	631	0.40	2,373	26.6%	155
Corvane Quillbury (2163)	Quillbury Grid Operator	538	0.39	2,338	23.0%	183
Corvane Kestrel Point (2402)	Kestrel Point Grid Authority	680	0.39	2,462	27.6%	225
Corvane Faraview (2317)	Faraview Power Administration	474	0.39	1,976	24.0%	122
Corvane Ostlemere (2356)	Ostlemere Electric Grid	864	0.39	2,862	30.2%	170

Sources: Electricity Data Commission hourly operating data and 2025 generation and emission rates (release 1.0); Lumen Ridge Solar Engineering array model runs on 2025 weather; Corvane accounts payable. Stores in order of grid CO2 avoided.

How the avoided carbon dioxide was measured

For each area I put the agency's hourly operating data and each store's modelled array output on one clock (the operating data name each hour by its end in UTC, the model by its start in local standard time) and estimated the area's marginal emission rate in each season (December to February, March to May, June to August, September to November) and hour of the local day: the hour-to-hour change in the area's fossil carbon dioxide (its hourly coal, gas and oil generation at the agency's 2025 output emission rates for each fuel) against the hour-to-hour change in its net generation, which with interchange scheduled a day ahead is the change its own generators make to follow the load. Each store's figure is that rate weighted by its array's output, hour by hour. Month-by-hour bins, the hour of day alone, or changes in demand or fossil generation in place of net generation give the same figures to two decimals.

Use and price from the invoices

Metered use is the kilowatt-hours each store's meters recorded in 2025. Each delivery utility's own invoices carry them (its distribution energy quantity, once per meter and billing cycle; Tennaford bills it in declining blocks, which are summed) until the store's main account moved to Halworth Utility Payment Services LLC in the spring of 2025; the garden-centre accounts at Quillbury and Kestrel Point stayed on direct payment all year. The service's lines carry only what it paid, so for the rest of the year the use comes from Ardley Retail Energy LLC, which bills each meter on the utility's read cycles. Its first term billed the meter's kilowatt-hours grossed up to the wholesale delivery point, 1.0365 to 1.0623 times the meters' readings in every cycle both bill, so those bills are divided by each meter's factor; its new term, from the first read on or after 1 July, bills the meter's kilowatt-hours with the renewable standard's cost in the price, as the garden-centre meters show, so those bills are taken as billed.

The price is everything Corvane paid for each store's electricity for 2025 service over that use, each bill counted by its days in 2025: the utilities' invoices, the service's payments and the supplier's invoices. Seven stores' first bill after the move was paid twice, by accounts payable and by the service; the utility carried the overpayment as a credit and the service paid that much less on the next bills, so every payment is counted as made. Halvane Gas & Electric's gas charges are left out. Each array is sized below its store's daytime load (charter 3.3), so the share supplied is its 2025 output over the store's use.

Kestrel Point's area filed its data under its former code, KPLP, until 30 June 2025. The operator filed a few hours of demand from each hour's net generation and interchange after telemetry gaps; leaving them out moves no figure.

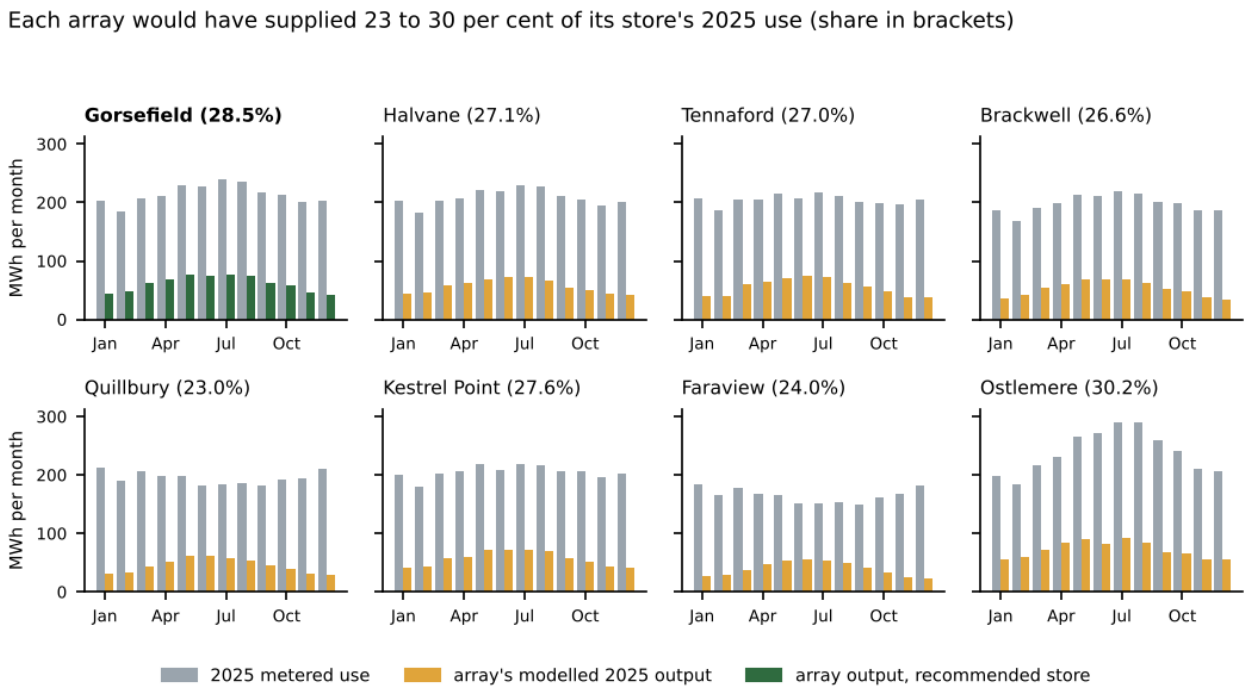


Figure 1. Each store's 2025 metered use by calendar month against its array's modelled 2025 output.

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